

Honours Programme in

Data Science for Emerging Analytics

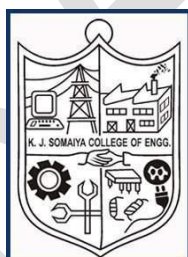
(Offered by Department of Electronics and Telecommunication Engineering)

From

Academic Year 2024-25

(Revision-2)

Approved by Academic Council dated 4th July 2024



K J Somaiya College of Engineering, Mumbai-77

(A Constituent College of Somaiya Vidyavihar University)

Offered by Department of Electronics and Telecommunication Engineering

In our contemporary, data-centric society, the field of Data Science stands as a fundamental pillar, offering a robust framework for addressing challenges across a multitude of sectors including science, engineering, medicine, marketing, finance, and beyond. By leveraging advanced analytics and processing techniques, Data Science enables the extraction of valuable insights from vast and diverse datasets, spanning text, audio, video, and other formats and transforms raw data into meaningful insights, driving informed decision-making and driving innovation across industries.

Data visualization represents one of the fundamental tools of modern data science. Data visualization is useful for data cleaning, exploring data structure, detecting outliers , identifying trends , spotting local patterns, evaluating modeling output, and presenting results. It is essential for exploratory data analysis and data mining to check data quality and to help analysts become familiar with the structure and features of the data before them. Data visualization helps to choose appropriate model for modelling data.

Deep learning is deemed as one of the most important revolutions in computer science in the past decade. It has been demonstrated to be extremely effective in learning and prediction tasks, such as computer vision, natural language processing, robotics and so on. The development in the field is very fast and hence mandates study of recent advances in deep learning.

Recently the field of Data Science is revolutionized by Large Language Models (LLMs) which have ushered the world into a new era with rich new capabilities for wide-ranging application domains. They form the basis of state-of-art systems and become ubiquitous in solving a wide range of natural language and image understanding and generation tasks. At the same time, there is a need to understand their capabilities and functioning for their application in diverse domains and for further betterment.

With the availability of Data Science Tools and Techniques, it is essential to explore their application to solve a variety of challenges in diverse domains. Particularly important are domains associated with industry

A “Data Science for Emerging Analytics” program is meticulously crafted to equip students with a comprehensive skill set tailored to meet the demands of today's data-driven landscape. Through a combination of theoretical coursework and hands-on practical experience, students are immersed in the intricacies of data analysis, Deep Learning, and data visualization

techniques. The course includes application of modern Data Science tools for Social Media and Web data analytics as well as Marketing analytics. This holistic approach fosters the development of critical problem-solving skills and computational acumen, empowering students to address real-world challenges with confidence and ingenuity.

Objectives:

- The fundamental goal of data analytics is to extract meaningful insights and valuable knowledge from vast and intricate datasets. These insights serve several purposes:
- Informed Decision-Making
- Predictive Modelling
- Competitive Advantage
- Performance Improvement

Learning Outcomes of the Honours Degree Programme:

At the successful completion of this programme an engineering graduates will be able to

- Apply principles of Data Science for analytics to diverse problems.
- Demonstrate the use of various tools in the domain of Data Science for data visualization.
- Discuss ethical practices related to data-driven decision-making.
- Implement solutions to data analysis problems using latest tools and technologies.

Assessment Methods: Evaluation is done by a variety of tools including Open book tests, MCQs (multiple choice questions), Study of research papers, Internal Assessment tools and End Semester Examinations etc. Mini-Projects are offered in courses also to encourage project based learning among students.

Acronyms used in syllabus document	
Acronym	Definition
CA	Continuous Assessment
ESE	End Semester Exam
IA	Internal Assessment
O	Oral
P	Practical
P&O	Practical and Oral

TH	Theory
TUT	Tutorial
TW	Term work
ISE	In-semester Examination
CO	Course Outcome

Acronyms used in Course code e.g. 116HxxC301

Position of Digit	Acronym	Definition
1	2	SUV 2023 Second Revision
2	16	KJSCE
3	H	Honour Degree Program
4	04 (XX)	Software Project Management
5	C	Core Course
	L	Laboratory Course
	T	Tutorial
	P	Project Based Course
6	1/2/3/4	Semester Number
7	01/02/03--	Course Number

Proposed Credit Scheme

Course Code	Course Name	Teaching Scheme (Hrs.) TH – P – TUT	Total (Hrs.)	Credits Assigned TH – P – TUT	Total Credits	Suggested semester of Honours' degree
216h27C301	Data Visualization	3 – 0 – 0	03	3 – 0 – 0	03	III
216h27L301	Data Visualization Laboratory	0 – 2 – 0	02	0 – 1 – 0	01	III
216h27C401	Advanced Data Analytics	3 – 0 – 0	03	3 – 0 – 0	03	IV
216h27L401	Advanced Data Analytics Laboratory	0 – 2 – 0	02	0 – 1 – 0	01	IV
216h27C501	Data Mining and Warehousing	3 – 0 – 0	03	3 – 0 – 0	03	V
216h27L501	Data Mining and Warehousing Laboratory	0 – 2 – 0	02	0 – 1 – 0	01	V
216h27C601	Deep Learning	3 – 0 – 0	03	3 – 0 – 0	03	VI
216h27L601	Deep Learning Laboratory	0 – 2 – 0	02	0 – 1 – 0	01	VI
216h27L701	Domain Specific Analytical Project	1 – 2 – 0	03	0 – 2 – 0	02	VII
	Total	13 – 10 – 0	23	12 – 6 – 0	18	

Proposed Examination Scheme

Course Code	Course Name	Examination Scheme				
		Marks				
		CA		ESE	Lab/ Tut CA	Total
		ISE	IA			
216h27C301	Data Visualization	30	20	50	--	100
216h27L301	Data Visualization Laboratory	-	-	-	50	50
216h27C401	Advanced Data Analytic	30	20	50	--	100
216h27L401	Advanced Data Analytic Laboratory	-	-	-	50	50
216h27C501	Data Mining and Warehousing	30	20	50	--	100
216h27L501	Data Mining and Warehousing Laboratory	-	-	-	50	50
216h27C601	Deep Learning	30	20	50	--	100
216h27L601	Deep Learning Laboratory	-	-	-	50	50
216h27L701	Domain Specific Analytical Project	-	-	-	75	75
Total		120	80	200	275	675

Course Code	Name of the Course				
216h27C301	Data Visualization				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course prerequisites (if any):

Basics of statistics, database and data analysis

Course Objectives

To provide students with the foundations necessary for understanding and extending the current state of the art in data visualization, to gain an understanding of the key techniques and theory used in visualization, including data models, graphical perception and techniques for visual encoding and interaction, to plan for data-based storytelling through charts, maps, and diagrams, to use visualization tools to transform quantitative information to visual representation, and to gain practical experience building and evaluating visualizations.

Course Outcomes

At the end of successful completion of the course the student will be able to

CO1	Learn how to locate and download datasets, extract insights from that data and present their findings in a variety of different formats
CO2	Detect and understand the stories within datasets and its applications.
CO3	Apply data visualization best practices
CO4	Design static charts, interactive Dashboards

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction data visualization		10	CO1 CO2
	1.1	Introduction to Data Acquisition – Applications – Process- Data Extraction- Data Cleaning and Annotation- Data Integration -Data Reduction- Data Transformation – Visualization-Introduction - Terminology		
	1.2	Basic Charts and Plots- Multivariate Data Visualization- Data Visualization Techniques– Pixel-Oriented Visualization Techniques- Geometric Projection Visualization Techniques- Icon-Based Visualization Techniques- Hierarchical Visualization Techniques .Visualizing Complex Data and Relations The Bar Chart,The Line Chart,The Pie and Donut Charts The Scatter Plot,The Packed Bubble Chart, Clustered bar charts, dot plots, connected dot plots, pictograms		
2	Data Visualization Tools–		06	CO 3
	2.1	Rank Analysis Tools- Trend Analysis Tools- Multivariate Analysis Tools- Distribution Analysis Tools- Correlation Analysis Tools- Geographical Analysis Tools.		
3	Storytelling and Multivariate displays		10	CO4
	3.1	Storytelling Multivariate displays The Science of Storytelling,The Power of Stories Context in Action,Exploratory versus Explanatory Analysis,Structuring Stories,Audience Analysis for Storytelling,Steps to Visual Data Storytelling The Important Role of Feedback		
	3.2	Graphical Perception		
	# Multivariate displays			
4	Geospatial displays		09	CO4
	4.1	The Heat Map Maps:Connecting to Geographic Data,Assigning Geographic Roles, Creating Geographic Hierarchies Proportional Symbol Maps, Choropleth Map		
5	Dashboards, interactive and animated displays		10	CO4
	5.1	Visual Design Building Blocks Color: Stepped Color, Reversed Color Color Effects: Opacity, Mark Borders, Mark Halos The Truth about Red and Green Lines: Formatting Grid Lines, Zero Lines, and Drop		

	Lines Formatting Borders,Formatting, Shading, and Banding Shapes Shape Marks Card, Custom Shapes Timelines,Bar-in-Bar Charts ,Bubble Chart ,Likert Visualizations,Lollipop Charts,Word Clouds,Create dashboard,Working with dashboard ,Publishing through dashboard		
	#case study, Publishing dashboard over mobile devices		
Total		45	

Students should prepare all Self Learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and Laboratory Experiments.

Reference Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Sosulski, K.	<i>Data Visualization Made Simple: Insights into Becoming Visual</i>	New York: Routledge.	First edition, 2018
2.	Lindy Ryan	<i>Visual Data Storytelling with Tableau</i>	Pearson Education	First edition, 2018
3.	Kristen Sosulski	Data Visualization made simple	New York: Routledge	First edition, 2019
4.	Cole Nussbaumer Knaflic	Storytelling with Data	Wiley	First edition, 2015

Course Code	Name of the Course				
216h27L301	Data Visualization Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		50	-	-	-

*Distribution of marks:

Instructions:	• LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. • LAB CA also included oral / Evaluation based on entire syllabus (25 marks). • LAB CA oral schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
----------------------	---

Course Code	Name of the Course				
216h27C401	Advanced Data Analytics				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		-	20	30	50

Course prerequisites (if any):

Basics of statistics, database and data analysis

Course Objectives

To provide students with the foundations necessary for understanding and extending the current state of the art in data visualization, to gain an understanding of the key techniques and theory used in visualization, including data models, graphical perception and techniques for visual encoding and interaction, to plan for data-based storytelling through charts, maps, and diagrams, to use visualization tools to transform quantitative information to visual representation, and to gain practical experience building and evaluating visualizations.

Course Outcomes

At the end of successful completion of the course the student will be able to

CO1	Explore the various forms of electronic health care information. And techniques adopted to analyse health care data.
CO2	Apply data-driven, machine learning approaches for improvements in farming techniques
CO3	Design modeling, aggregating and knowledge representation of semantic web
CO4	Collect, analyse, and interpret data, making informed decisions based on quantitative insights.

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction to Exploratory Data Analysis		10	CO1 CO2
	1.1	Introduction to Exploratory Data Analysis (EDA) steps , Purpose of EDA ,Advantages of EDA ,Applications of EDA		
	1.2	Framework for Scientific Exploration, Case study, Apply the EDA framework on a real-world dataset. Generate insights and create a story around them.		
2	Time Series Data Analysis		09	CO 3
	2.1	Overview of time series data, Decomposition of time series: trend, seasonality, and residuals, Additive and multiplicative models, Moving averages and smoothing techniques		
	2.2	Autoregressive (AR) and Moving Average (MA) Models: Autoregressive (AR) models, Moving Average (MA) models, Autoregressive Moving Average (ARMA) models, Estimation and model diagnostics		
3	Autoregressive Integrated Moving Average (ARIMA) Models		10	CO4
	3.1	Introduction to ARIMA models, Identifying the order of ARIMA models (p, d, q), Model fitting, selection, and diagnostics, Seasonal ARIMA (SARIMA) models		
	3.2	Simple exponential smoothing, Holt-Winters seasonal method, Introduction to state-space models, Kalman filter and its applications		
4	Business Analytics		09	CO4
	4.1	Advanced Excel for Analytics: Data Cleaning and Processing, Vlookup, Pivot table, Pivot Charts and Dashboards, Date functions, Conditional Formatting and Data Validation, VBA Functions and Procedures		
	4.2	Power BI: Concepts of Business intelligence (BI), Relevance of BI in application to analytics, industry and different domains, Report Generation and Dashboard creation using Power BI.		

5	GIS Data analytics		7	
	5.1	Perspective of spatial data science: Business, Technology and Data. Technologies for GIS Data analysis and application. Introduction to open source software tools: QGIS/Geospark-R/Saga		
Total			45	

Reference Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Sosulski, K.	<i>Data Visualization Made Simple: Insights into Becoming Visual</i>	New York: Routledge.	First edition, 2018
2.	Tarek A. Atwan	<i>Time Series Analysis with Python Cookbook</i>	Packt Publishing	Second edition, 2019
3.	By U Dinesh Kumar	<i>Business Analytics: The Science of Data - Driven Decision Making</i>	Willy	Second edition, 2021
4.	Deepankar Chakrabarti, Suchandra Choudhury	An Introduction to Geographic Information Technology	I K International Publishing House Pvt. Ltd	2013
5.	Errin O'Connor	Microsoft Power Bi Dashboards Step By Step	Pearson Education	2020

DRAFT

Course Code	Name of the Course				
216h27L401	Advanced Data Analytics Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		50	-	-	-

*Distribution of marks:

Instructions:	• LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. • LAB CA also included oral / Evaluation based on entire syllabus (25 marks). • LAB CA oral schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
----------------------	---

Course Code	Name of the Course				
216h27C501	Data Mining and Warehousing				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course pre-requisites:

- Knowledge of analytical tools

Course Objectives:

- To teach principles, concepts and applications of data warehousing and data mining
- To introduce the task of data mining as an important phase of knowledge recovery process
- To inculcate Conceptual, Logical, and Physical design of Data Warehouses OLAP applications and OLAP deployment
- To inculcate fundamental concepts that provide the foundation of data mining
- To introduce classification, association, and clustering techniques to extract patterns from data.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1. Learn concept and architecture of Data Warehousing
 CO2. Explain data warehouse design and OLAP
 CO3. Implement data preprocessing operations and association rule mining
 CO4. Implement machine learning algorithms for data classification and clustering to extract patterns from data
 CO5. Explore recent trends in data mining like web mining, spatial data mining

Detailed Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Data Warehousing- Introduction and Architecture			
	1.1	Overview of Data Warehouse: Definition, working and types of data warehouse; general stages of a data warehouse; components of data warehouse; goals of data warehouse; characteristics of data warehouse; data warehouse users; how users query the data warehouse; applications of data warehouse	7	CO1, CO2
	1.2	Definition of Data Warehouse Architecture; Data Warehouse Design Process; A Three Tier Data Warehouse Architecture; Data warehouse Back-End Tools and Utilities- Extract, Transform, Cleansing, Loading		
2	Data Warehouse Design and OLAP			
	2.1	Data Warehouse Design: Logical Versus Physical Design of Data Warehouses, Data Warehouse Design Methodologies- Bottom-up, Top-down, Hybrid; Developing a Data Warehouse: identify and gather requirements; design the dimensional model-Data Cube, Star, Snowflakes, Fact Constellations; DMQL; Concept Hierarchies; OLAP operations on multidimensional data	8	CO2, CO3
	2.2	Data Warehouse and OLAP: Meaning and benefits of OLAP, comparison of OLTP and OLAP ; From on-line analytical processing to on-line analytical mining, Architecture for on-line analytical mining		
3	Data Mining- Introduction and Association Rule Mining			
	3.1	Data Mining: Data mining process, Data mining Functionalities; Data Preprocessing – Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation; Architecture Of A Typical Data Mining Systems; Classification Of Data Mining Systems	10	CO3
	3.2	Association Rule Mining: Efficient and Scalable Frequent Item set Mining Methods, Market basket analysis, Mining Various Kinds of Association Rules,		

		Association Mining to Correlation Analysis, Constraint-Based Association Mining		
4	Data Mining – Classification and Clustering techniques			
	4.1	Classification : Basic classification task, Binary and Multiclass classification, train-validation-test data sets, Metrics for evaluation of classification performance, Classification using Machine Learning algorithms – Logistic regression, k-nearest neighbor, Decision Tree, Model selection, Cross validation, Ensemble methods to improve classification performance(Bagging, Boosting, Stacking)	14	CO4, CO5
	4.2	Clustering : Basic issues, Clustering using partitioning methods, Hierarchical methods, Density based methods, Grid based methods and model based methods, Algorithms for outlier analysis		
5	Advanced Mining			
	5.1	Spatial Data mining, Text Data mining, Relational Databases mining, Transaction databases mining	6	CO5
	5.2	Web mining, Web Content mining, Web Structure mining, Web Usage mining, Multimedia Data mining		
Total			45	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Jiawei Han, Micheline Kamber	Data Mining Concepts and Techniques	Elsevier, U.S.A.	3 rd edition, 2012
2.	Pualraj Ponniah	Data Ware Housing Fundamentals: A Comprehensive Guide for IT Professionals	Wiley, U.S.A.	2 nd edition, 2001
3.	Pang-Ning Tan, Vipin Kumar, Michael Steinbanch	Introduction to Data Mining	Pearson Education, U.S.A.	2 nd edition, 2018
4.	Alex Berson and Stephen J. Smith	Data Warehousing, Data Mining & OLAP	Tata McGraw Hill, India	2 nd edition, 2004
5.	M.H. Dunham	Data Mining Introductory and Advanced Topics	Pearson Education, India	1 st edition, 2006

Course Code	Name of the Course				
216h27L501	Data Mining and Warehousing Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	-	-	-	50

*Distribution of marks:

Instructions:	• LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. • LAB CA also included oral / Evaluation based on entire syllabus (25 marks). • LAB CA oral schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
----------------------	---

Course Code	Name of the Course				
216h27C601	Deep learning				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	150

Course pre-requisites:

- Mathematical background in calculus, linear algebra, and probability & statistics, as well as prior coursework in machine learning and programming experience in Python

Course Objectives:

Deep learning is a sub-field of machine learning that focuses on learning complex, hierarchical feature representations from raw data. The dominant method for achieving this, artificial neural networks, has revolutionized the processing of data (e.g. images, videos, text, and audio) as well as decision-making tasks. Its success has enabled a tremendous amount of practical commercial applications and has had a significant impact on society.

Course Outcomes (CO):
At the end of successful completion of the course the student will be able to

- CO1. Describe the major differences between deep learning and classical machine learning algorithms
- CO2. Explain the fundamental methods involved in deep learning, including the underlying optimization concepts (gradient descent and backpropagation), typical modules they consist of, and how they can be combined to solve real-world problems.
- CO3. Differentiate between the major types of neural network architectures (multi-layered perceptrons, convolutional neural networks, recurrent neural networks, etc.) and what types of problems each is appropriate for.
- CO4. Select or design neural network architectures for new data problems based on their requirements and problem characteristics and analyze their performance.

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction		6	CO1 CO2
	1.1	History and success stories of Deep Learning; Difference between Machine Learning and Deep Learning; McCulloch Pitts Neuron, Thresholding Logic, Perceptron Learning Algorithm; Perceptron with sigmoidal neuron, Gradient Descent(GD) algorithm, Multilayer Perceptron (MLP), Backpropagation error algorithm; activation functions; loss functions		
	1.2	Deep neural networks (DNNs): Difficulty of training DNNs, Greedy layerwise training, Optimization for training DNNs, Momentum Based GD,, Stochastic GD, Minibatch GD, RMSProp, Adam, learning rate schedulers; weight initialization methods		
2	Convolutional neural networks(CNNs)		10	CO 2, CO3, CO4
	2.1	Basic concepts of Convolutional Neural Networks , Convolution and pooling operations; Fully connected layers, output layer; Various activation functions; Deep CNN; Training a CNN, weights initialization, batch normalization, hyperparameter optimization, transfer learning; visualizing CNNs; learning famous CNN architectures - AlexNet, InceptionNet, VGG, GoogleNet, ResNet, MobileNet-v1, MobileNet-v2, EfficientNet		
	2.2	Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Dropout		
3	Recurrent Neural Networks(RNNs)		10	CO2, CO3
	3.1	RNNs, Sequence modeling using RNNs, word embedding for text data, Back propagation through time, Long Short Term Memory (LSTM), Bidirectional LSTMs; Exploding and Vanishing gradient, Truncated Back Propagation Through Time, gradient clipping; Deep RNNs; Encoder-Decoder models using RNNs		
4	Deep Unsupervised Learning		10	CO2, CO3
	4.1	Deep Unsupervised Learning, Autoencoders,		
	4.2	Generative Models: Variational Autoencoders, Generative Adversarial Networks, Diffusion Model ; Encoder-Decoder models with attention mechanism		
5	Advanced Topics and Applications		9	CO4
	5.1	Transformers: Multi-headed Self Attention, Cross Attention		

	5.2	Applications of: MLP, CNN, RNN, Encoder-Decoder models with and without attention, Transformers. (Currently relevant applications to be covered)		
Total			45	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Ian Goodfellow and Yoshua Bengio and Aaron Courville	Deep Learning	MIT Press, U.S.A.	2016
2	Aston Zhang, Zack C. Lipton, Mu Li, and Alexander Smola	Dive into Deep Learning	Cambridge University Press, U.K.	1 st edition, 2023
3	Francois Chollet	Deep Learning with Python	Manning, U.S.A.	2 nd edition, 2021
4	Charu C. Aggarwal	Neural Networks and Deep Learning: A Textbook	Springer, Germany	1 st edition, 2018

Course Code	Name of the Course				
216h27L601	Deep learning Laboratory				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	-	02	-	02	
Credits Assigned	-	01	--	01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	-	-	-	50

*Distribution of marks:

Instructions:	• LAB CA includes journal / written record of experiments (25 marks each). A minimum of 8-10 experiments will be performed in a semester. • LAB CA also included oral / Evaluation based on entire syllabus (25 marks). • LAB CA oral schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.
----------------------	---

Course Code	Name of the Course				
216h27L701	Domain Specific Analytical Project				
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total	
	01	02	--	03	
Credits Assigned	--	02	--	02	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
		75	-	-	--

Course prerequisites : Probability and Statistics, machine learning models, data visualization

Course Objectives: Data analytics is the process of storing, organizing, and analyzing raw data to answer questions or gain important insights. Data analytics is integral to business because it allows leadership to create evidence-based strategy, understand customers to better target marketing initiatives, and increase overall productivity. Companies that take advantage of data analytics reap a competitive advantage because they are able to make faster changes that increase revenue, lower costs, and spur innovation. In today's digital world, the ability to make data-driven decisions and create strategy informed by analysis in related domains including Marketing, Finance, Supply Chain and Human Resources is central to successful leadership in any industry.

Course Outcomes

At the end of successful completion of the course the student will be able to

CO1	Learn and apply data science tools and techniques for Healthcare Data Analytics
CO2	Learn and apply data science tools and techniques for Energy and Environmental Analytics
CO3	Learn and apply data science tools and techniques for Social Media and Marketing Analytics
CO4	Learn and apply data science tools and techniques for Manufacturing and Supply Chain Analytics

Module No.	Unit No.	Details	Hrs.	CO
1	Healthcare Data Analytics		4	CO1
	1.1	Introduction: Introduction to Healthcare Data Analytics- Electronic Health Records– Components of EHR- Coding Systems- Benefits of EHR- Barrier to Adopting EHRChallenges- Phenotyping Algorithms.		
	1.2	Analysis: Biomedical Image Analysis- Mining of Sensor Data in Healthcare- Biomedical Signal Analysis- Genomic Data Analysis for Personalized Medicine.		
2	Energy and Environmental Analytics		4	CO 2
	2.1	Introduction to Energy and Environmental Analytics, Overview of energy systems and environmental challenges Importance of data analytics in addressing energy and environmental issues Data Sources and Collection: Data collection from smart grids and smart meters		
		IoT and Smart Sensors for Environmental Monitoring: Design and deployment of IoT systems, Use of drones and UAVs for environmental data collection, Case studies: Smart water management, air quality monitoring Sustainability and Renewable Energy Analytics: Data-driven approaches for assessing sustainability, Analytics in renewable energy integration (solar, wind, hydro), Case studies: Carbon footprint analysis, sustainable resource management		
3	Social Media and Marketing Analytics		3	CO3
	3.1	Social Network analysis: Development of Social Network Analysis – Key concepts and measures in network analysis Ontology and their role in the Semantic Web: Ontology-based knowledge Representation – Ontology languages for the Semantic Web: Resource Description Framework – Web Ontology Language		
	3.2	– Modelling and aggregating social network data: State-of-the-art in network data representation – Ontological representation of social individuals – Ontological		

		representation of social relationships – Aggregating and reasoning with social network data – Advanced representations.		
4	Supply Chain Analytics			
	4.1	Introduction to Supply Chain Management. Data Collection and Management. Forecasting and Demand Planning. Inventory Management. Network Design and Optimization	4	CO4
Total			15	

Recommended Books: Will be made available by the concerned faculty at the time of teaching.

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Prof. Dr. R Gopal	Essentials Of Healthcare Analytics	Notion Press	2024
2	Ralph R.B. Von Frese	Basic Environmental Data Analysis for Scientist and Engineers	CRC Press	2019
3	Ganis/Kohirkar	Social Media Analytics	Pearson Education India	2016
4	T. A. S. Vijayaraghavan	Supply Chain Analytics	Wiley	2021

*In addition to printed books, faculty can suggest (authentic) URLs or e-books, e-contents etc.

*Distribution of marks:

Instructions:	- LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge - There will be continuous assessment of the experiments on each module and progress report of the projects needs to be maintained
----------------------	---

	by students. • The final presentation will be based on a demonstration of the project in front of a committee of examiners. Students are expected to publish technical paper based on the project.
--	---

DRAFT